Magma & LMFDB Mini-Course

Hunt 2: Number Fields and Kronecker–Weber

Eran Assaf

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We will find a non-negative integer n. Each of the four core problems below has a single non-negative integer answer n_i. The bonus is harder and does not count toward the sum. Set

$$n \;=\; n_1 + n_2 + n_3 + n_4.$$

Solutions in Magma or Sage are both welcome.

Problem 2.1. The smallest absolute discriminant of a cubic number field is 23 (the running cubic $\mathbb{Q}[x]/(x^3 - x - 1)$). Let n_1 be the smallest absolute discriminant of a *totally real* cubic number field.

Problem 2.2. Let n_2 be the number of fundamental discriminants $-D$ with $0 < D \leq 1000$ for which the imaginary quadratic field $\mathbb{Q}(\sqrt{-D})$ has class number exactly 3.

Problem 2.3. By Kronecker–Weber, every finite abelian extension of $\mathbb{Q}$ is contained in a cyclotomic field $\mathbb{Q}(\zeta_n)$ for some n; the smallest such n is the *conductor*. Let n_3 be the conductor of the compositum $\mathbb{Q}(i, \sqrt{-23})$.

Problem 2.4. Let $L = \mathbb{Q}(\sqrt{-23})$ and let H be the Hilbert class field of L. Let n_4 be the smallest prime p that splits completely in $H/\mathbb{Q}$.

Bonus 2 (Cohen–Lenstra, empirical). The Cohen–Lenstra heuristic predicts that the proportion of imaginary quadratic fields $\mathbb{Q}(\sqrt{-D})$ (with $-D$ a fundamental discriminant) whose odd-part class group is trivial is

$$\prod_{k=1}^{\infty} (1 - 2^{-k}) \approx 0.2887...$$

Equivalently: $h(\mathbb{Q}(\sqrt{-D}))$ is odd for approximately 28.88% of imaginary quadratic fields.

(a) Verify this empirically. Compute the proportion of imaginary quadratic fields with $|\operatorname{disc}| \leq 10^4$ (resp. $\leq 10^5$) whose class number is odd. Compare with the predicted constant.

(b) Among the same fields (with odd class number), compute the proportion whose class group is cyclic. Cohen–Lenstra predicts the proportion to approach

$$\zeta^{(2)}(6)^{-1} \prod_{k=4}^{\infty} \zeta^{(2)}(k)^{-1} \approx 0.9775...,$$

where $\zeta^{(2)}(s) = (1 - 2^{-s})\zeta(s)$. How well does the data agree?

(c) Plot the running proportion as a function of the discriminant bound. Does it appear to converge?